Longitudinal modes of metallic nanostructures

This set of notes is on the paper Nonlocal Response of Metallic Nanospheres Probed by Light, Electrons, and Atoms (https://pubs.acs.org/doi/10.1021/nn406153k) by **Christensen et al**

As done in the supplement of the paper, we seek longitudinal plasma solutions ($\mathbf{J}(\mathbf{r},\omega)=\nabla\Psi(\mathbf{r},\omega)$) in a metallic sphere. We note that such longitudinal modes must satisfy the equation (using the notation of the paper):

$$(\nabla^2+k_{NL}^2)(\nabla\cdot\mathbf{J}(\mathbf{r},\omega))=0\rightarrow(\nabla^2+k_{NL}^2)(\nabla^2\Psi(\mathbf{r},\omega))=0,$$

where k_{NL} is defined in the paper and will be re-derived below. By requiring that $\Psi(\mathbf{r},\omega)$ is a solution to the scalar Helmholtz equation, we obtain $\Psi(\mathbf{r},\omega)=j_l(k_{NL}r)Y_{lm}(\theta,\phi)$. Furthermore, since $\mathbf{e}_r\cdot\mathbf{J}(\mathbf{r},\omega)\propto\partial_r j_l(k_{NL}r)$ (and since we require that the normal component of the current vanish at the sphere's boundary, $r=R$), we obtain the condition $j_l'(k_{NL}R)=0$. Furthermore, note that k_{NL} may be readily found through Maxwell's equations combined with the hydrodynamic Ohm's law :

$$\frac{\beta_F^2}{\omega(\omega+i\eta)}\nabla(\nabla\cdot\mathbf{J}(\mathbf{r},\omega))+\mathbf{J}(\mathbf{r},\omega)=\frac{i\omega_p^2\varepsilon_0}{\omega+i\eta}\mathbf{E}(\mathbf{r},\omega)\rightarrow$$
$$\frac{\beta_F^2}{\omega(\omega+i\eta)}\nabla^2(\nabla\cdot\mathbf{J}(\mathbf{r},\omega))+\nabla\cdot\mathbf{J}(\mathbf{r},\omega)-\frac{i\omega_p^2\varepsilon_0}{\omega+i\eta}\nabla\cdot\mathbf{E}(\mathbf{r},\omega)=0\rightarrow$$
$$\frac{\beta_F^2}{\omega(\omega+i\eta)}\nabla^2(\nabla\cdot\mathbf{J}(\mathbf{r},\omega))+\left[1-\frac{\omega_p^2}{\omega(\omega+i\eta)}\right]\nabla\cdot\mathbf{J}(\mathbf{r},\omega)=0,$$

where β_F has velocity units (and is typically taken as $\sqrt{3/5}v_F$ for a three dimensional metal and where we used the following (which follows from Ampere's law):

$$0 = \nabla \cdot (\nabla \times \mathbf{H}(\mathbf{r}, \omega)) = -i\omega\varepsilon_0 \nabla \cdot \mathbf{E}(\mathbf{r}, \omega) + \nabla \cdot \mathbf{J}(\mathbf{r}, \omega) \rightarrow$$
$$\nabla \cdot \mathbf{E}(\mathbf{r}, \omega) = (-i/\omega\varepsilon_0)\nabla \cdot \mathbf{J}(\mathbf{r}, \omega))$$

Thus, we have:

$$k_{NL}^2 = \frac{1}{\beta_F^2}\left[\omega(\omega + i\eta) - \omega_p^2\right]$$

Therefore, for purely longitudinal resonances, we obtain the following dispersion relation for a given zero of the spherical Bessel function's derivative, w_v:

$$k_{NL}^2 = w_v^2/R^2 \rightarrow \omega(\omega + i\eta) = \omega_p^2 + \frac{\beta_F^2 w_v^2}{R^2}$$

This is **Equation S22** of the supplement of the paper as well **Equation 9** of the main text.